

Sensores dos dispositivos móveis

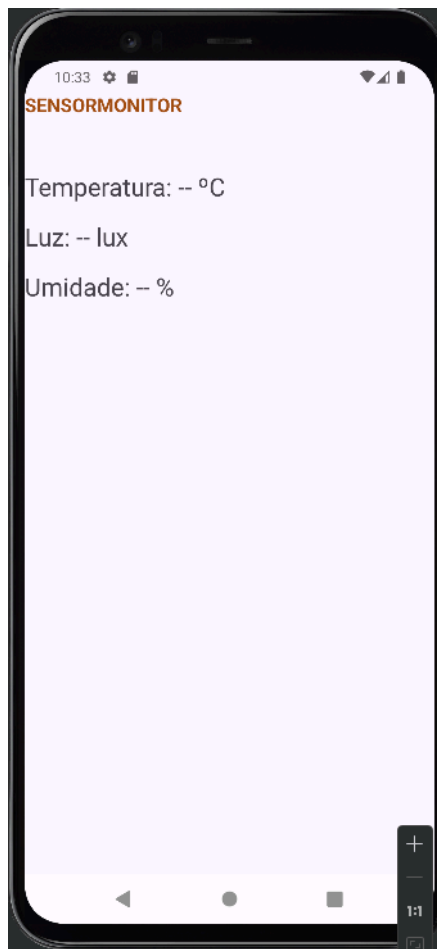
Temperatura, Luz e Umidade

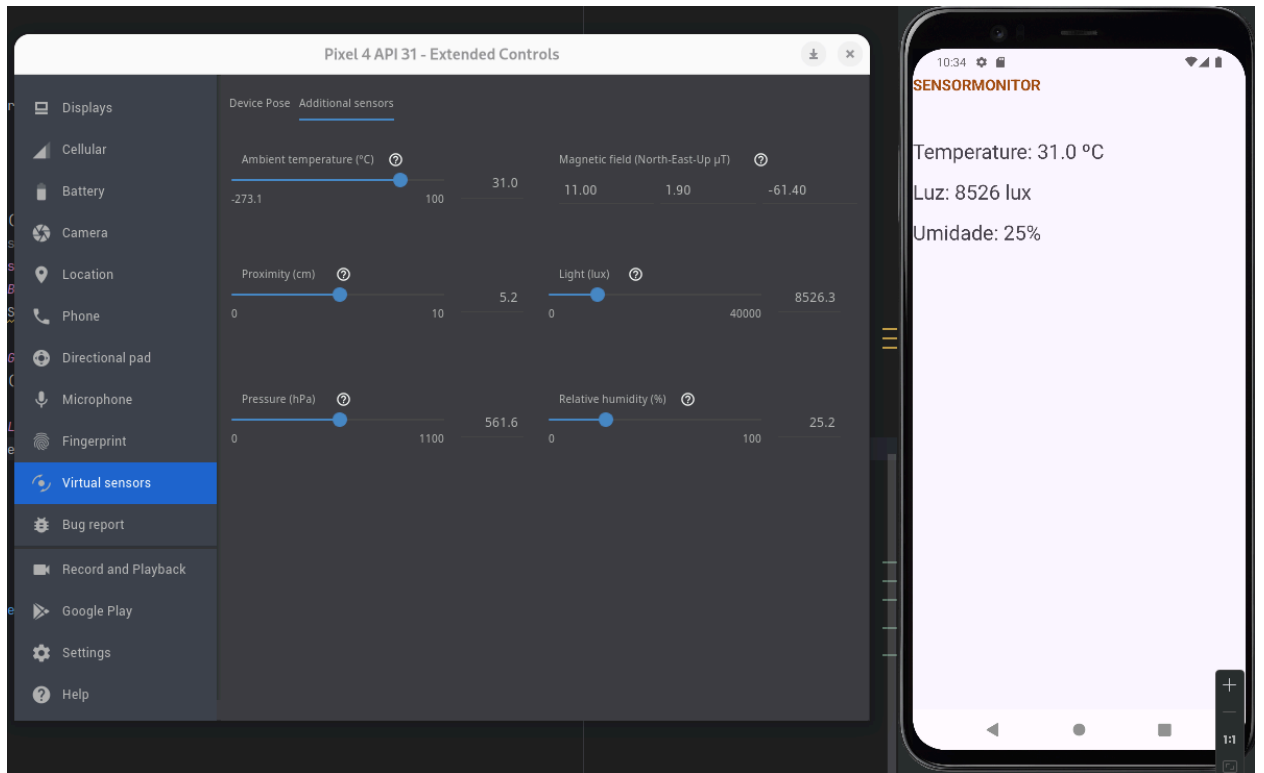
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Checkout de Presença Módulo 4.

Recursos de Hardware no Smartphone

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MainActivity

```
package com.example.aplicativodemonitoramentodesensoresandroid;

import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity implements SensorEventListener {

    private SensorManager sensorManager;

    private Sensor tempSensor, lightSensor, humiditySensor;

    private TextView tvTemp, tvLight, tvHumidity;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        EdgeToEdge.enable(this);
```

```

        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {

            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());

            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);

            return insets;

        });

        // Inicializa os TextViews

        tvTemp = findViewById(R.id.tvTemperature);

        tvLight = findViewById(R.id.tvLight);

        tvHumidity = findViewById(R.id.tvHumidity);

        // Configura o SensorManager

        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);

        // Pega os sensores e sua disponibilidade

        tempSensor =
sensorManager.getDefaultSensor(Sensor.TYPE_AMBIENT_TEMPERATURE);

        lightSensor = sensorManager.getDefaultSensor(Sensor.TYPE_LIGHT);

        humiditySensor =
sensorManager.getDefaultSensor(Sensor.TYPE_RELATIVE_HUMIDITY);

```

```

        // Tratamento de sensores

        if (tempSensor == null) tvTemp.setText("Sensor de temperaura não
disponível");

        if (lightSensor == null) tvLight.setText("Sensor de lux não disponível");

        if (humiditySensor == null) tvHumidity.setText("Sensor de humidade não
disponível");

    }

    @Override

    protected void onResume() {

        super.onResume();

        // Registra listeners com taxa de atualização

        if (tempSensor != null) sensorManager.registerListener(this, tempSensor,
SensorManager.SENSOR_DELAY_NORMAL);

        if (lightSensor != null) sensorManager.registerListener(this, lightSensor,
SensorManager.SENSOR_DELAY_NORMAL);

        if (humiditySensor != null) sensorManager.registerListener(this,
humiditySensor, SensorManager.SENSOR_DELAY_NORMAL);

    }

    @Override

    protected void onPause() {

        super.onPause();

        sensorManager.unregisterListener(this);

```

```

    }

    @Override

    public void onSensorChanged(SensorEvent sensorEvent) {

        // Atualiza os TextViews de acordo com o sensor

        switch (sensorEvent.sensor.getType()) {

            case Sensor.TYPE_AMBIENT_TEMPERATURE:

                tvTemp.setText(String.format("Temperature: %.1f °C",
sensorEvent.values[0]));

                break;

            case Sensor.TYPE_LIGHT:

                tvLight.setText(String.format("Luz: %.0f lux", sensorEvent.values[0]));

                break;

            case Sensor.TYPE_RELATIVE_HUMIDITY:

                tvHumidity.setText(String.format("Umidade: %.0f%%",
sensorEvent.values[0]));

                break;

        }

    }

    @Override

    public void onAccuracyChanged(Sensor sensor, int i) {

    }

}

```